Qn. Set Code-7 (SOT)

Semester: 4th
Programme: B.Tech
Branch: CSSE

SPRING END SEMESTER EXAMINATION-2024

4th Semester B.Tech

DESIGN & ANALYSIS OF ALGORITHMS

CS 2012

(For 2021 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions: [1 × 10]

(a) *$f(n) = \Theta(g(n))$ implies that.....*
Which of the following is true?
Give justification to your answer.

A. $0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n)$ v $n \geq n_0$, where c_1, c_2, and n_0 are +ve integers

B. $0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n)$ v $n \geq n_0$, for some value of c_1, c_2, and n_0 where they are +ve integers

C. $0 \leq c_1\, g(n) \leq f(n) \leq c_2 g(n)$ v $n \geq n_0$, for any value of c_1, c_2, and n_0 , where they are +ve integers

D. None of these

(b) Arrange the following functions in non-decreasing order of their growth.

$n^{2\log 2}$, $2^{0.5\log n}$, $2^{n\log n}$, $2^{\log n}$, $2^{\sqrt{n}\log n}$, $n!$, $3n^{\sqrt{n}}$

(c) Write the asymptotic tight bound of n log(n!).

(d) Consider the following function.

```
int KIIT( int x, int n)
{
if ( n==0)
return 1;
else if(n==1)
return x;
else if (n%2==0)
return KIIT( x * x, n/2);
else
return KIIT( x * x, n/2 ) * x ;
}
```

Find down the recurrence T(n) for the above function KIIT**(int x, int n)**.

(e) What are the similarities and dissimilarities between Divide-and-Conquer strategy and Dynamic-Programming strategy?

(f) Match the following Algorithms and its Complexities

(A) O (logn)	(a) Finding max-min in an array
(B) O (n)	(b) Heap-sort
(C) O (nlogn)	(c) Binary search
(D) O (n^2)	(d) Insertion sort

(g) An Undirected graph G has n nodes, its adjacency matrix is given by n×n square matrix whose diagonal elements are 0's and non-diagonal elements are 1's. Which of the following is true? Give justification to your answer.

(i) Graph G has no minimum spanning tree (MST)

(ii) Graph G has a unique MST of cost n-1

(iii) Graph G has multiple distinct MSTs, each of cost n-1

(iv) Graph G has multiple spanning trees different costs

(h) How many times MAX-HEAPIFY will be called if BUILD-MAX-HEAP is run on the given array elements (25, 17, 16, 12, 11, 9, 6, 4, 3, 2).

(i) Consider a Quick-sort algorithm that always selects $(n/5)^{th}$ smallest as the pivot element using O(n) time algorithm. What is the worst-case time complexity of the given Quick-sort algorithm?

(j) Given a big array, how to efficiently find $k^{'th}$ largest element in it?

SECTION-B

2. (a) State and explain master's method, and use the method to give tight asymptotic bounds for the recurrence [5]

$T(n) = 4T(n/2) + n^3$.

(b) Solve the recurrence using changing of variable or recursion-tree based method to determine a good asymptotic upper bound on the recurrence $T(n) = T(n/3) + n^{4/3}$ for $n > 3$ and $T(3) = 1$. [5]

3. (a) An array is initially sorted. When new elements are added, they are inserted at the end of the array and counted. Whenever the number of elements reached20, the array is resorted and counter is cleared. Which sorting algorithm would be good choice to use for resorting the array? Justify & Write the Algorithm and finds its time complexity. [5]

(b) Write a Quick-Sort algorithm that randomly selects an element as pivot element and derive the time complexity of this algorithm. [5]

4. (a) Write the algorithm to build a MAX-HEAP? Describe your Algorithm to build a MAX-HEAP from the array A = {6, 5, 8, 2, 1, 0, 3, 7, 4, 9, 6} in a step by step [5]

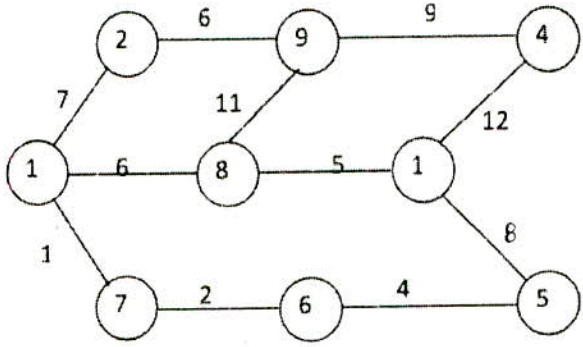
process. Derive the time complexity of building a MAX-HEAP.

(b) We are given an array of n elements, where first $2/3^{rd}$ of the array elements are in ascending order and rest of the array elements are in descending order. Write an Algorithm to sort the array in O(n) time. [5]

5. (a) Given 10 activities, A=< a1, a2,….,an > along with their start time (si) and finish time (fi) as Si=<2, 2, 4, 3, 7, 8, 8, 9, 11, 12 > and Fi=< 3, 5, 6, 7, 9, 11, 14, 18, 10, 15> and that requires the exclusive use of a common stage for scheduling these activities. Use an efficient method that computes a schedule with largest number of activities on that stage. [5]

(b) Find the optimal parenthesization of a Matrix-Chain product whose sequence of dimensions is < 5×10, 10×8, 8×7, 7×6, 6×5 > and write the algorithm for Matrix-Chain product. [5]

6. (a) Generate the minimum cost spanning tree from the following graph using Prim's method. [5]



(b) Define and differentiate between P, NP and NP-complete problems with examples. [5]

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